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PHD RESEARCH PROPOSAL DEFENSE — SEMESTER 2, 2025/2026

Enhanced Drug-Target Interaction Prediction Through Hybrid Deep Learning Approaches

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Presentation Outline

OVERVIEW

- 1. Motivation & Problem Statement
- 2. Research Questions and Objectives
- 3. Scope of the Research
- 4. Research Design — Three Pillars
- 5. Methodology: Data Collection & Preprocessing
- 6. Methodology: The H-GNN-RNN-DTI Architecture (mathematical formulation)
- 7. Methodology: Training Strategy
- 8. Methodology: Evaluation Protocol
- 9. Methodology: Interpretability Framework
- 10. Expected Outcomes, Timeline & Summary

Motivation — The Drug Discovery Bottleneck

- A single new chemistry-based drug costs ∼US$2.6 billion and ∼15 years to reach approval (Khojasteh et al., 2023)
- PubChem alone lists tens of millions of candidate compounds — exhaustive wet-lab testing against all targets is physically and economically infeasible
- Wet-lab assays and physics-based docking are accurate but slow, expensive, and require 3D target structures that are often unavailable
- Computational DTI prediction reframes discovery as a data-driven pattern-recognition problem — enabling virtual screening at scale
- Deep learning models drugs as molecular graphs and proteins as biological sequences — a natural fit for GNNs and RNNs respectively

Problem Statement — Four Persistent Gaps

- Weak cross-modal feature integration — GSRF-DTI, TTModel use static, late concatenation; no iterative drug↔protein interaction before prediction
- Limited multimodal representation learning — network/KG-level features substitute for direct GNN + RNN encoding of raw structure and sequence
- Poor cold-start generalization — models memorize seen drug/target patterns rather than learning transferable biochemical principles
- Black-box interpretability — Random Forest / KG-embedding backbones give no atom- or residue-level attribution for predictions
- Unaddressed class imbalance — confirmed interactions are vastly outnumbered by unlabeled/negative pairs, biasing training (Zhu et al. 2024, explicit future-work gap)

Why Existing Hybrid Models Fall Short

Model	Fusion Strategy	Key Limitation
GSRF-DTI (Zhu et al. 2024)	DeepWalk features → concatenation → Random Forest	No GNN/RNN encoders; static late fusion; no interpretability beyond feature importance
TTModel (Li et al. 2024)	Knowledge-graph + textual/type embeddings	Ignores molecular graph topology and raw protein sequence entirely
DTKGIN (Luo et al. 2024)	Intent-graph representation	Best cold-start baseline, but no dynamic attention fusion between structure and sequence

Research Questions

- RQ1 — How can a Gated Cross-Attention fusion mechanism effectively integrate GNN-based drug and RNN-based protein representations, and how does this improve DTI prediction over static concatenation?
- RQ2 — To what extent do pre-trained molecular/protein language embeddings combined with imbalance-aware learning improve generalization to unseen drugs, targets, and cross-dataset scenarios?
- RQ3 — Can an attention-enhanced hybrid model provide interpretable, biologically meaningful insight into key binding residues and pharmacophoric features?

Research Aim & Objectives

- Aim — Design an enhanced hybrid deep learning model combining GNNs and RNNs through a novel dynamic cross-modal fusion mechanism to improve DTI prediction, addressing the four identified gaps.
- O1 — Design and evaluate a hybrid model with an advanced cross-modal fusion mechanism, achieving statistically significant improvement over concatenation-based fusion baselines.
- O2 — Develop an imbalance-aware, foundation-model-pretrained representation learning strategy that measurably improves performance on unseen drugs, unseen targets, and cross-dataset transfer (cold-start).
- O3 — Implement and benchmark an attention-based interpretability layer producing quantitative and visual explanations, validated against known binding sites, to support SAR studies.

Scope of the Research

In Scope

- Binary DTI classification + continuous DTA regression (pKd/pKi/pIC50)
- SMILES → molecular graphs (RDKit); FASTA → ESM-2 embeddings
- Benchmarks: BindingDB, DAVIS, KIBA, DrugBank, ChEMBL, Luo_17, Yamanishi_08
- H-GAT + BiLSTM/BiGRU + Gated Cross-Attention architecture
- Cold-start evaluation (unseen drug / target / pair) + cross-dataset transfer
- Attention-based interpretability with case-study validation

Out of Scope

- Wet-lab / experimental validation of predicted interactions
- Non-small-molecule ligands (e.g., peptide or antibody therapeutics)
- Large-scale industrial / production deployment
- 3D structure-based docking simulation

Research Design — Three Pillars

- Quantitative, computational, experimental design — build, train, benchmark, ablate, interpret
- Pillar 1 ($\rightarrow$ O1): Dynamic cross-modal feature integration — H-GAT drug encoder + BiGRU protein encoder + Gated Cross-Attention
- Pillar 2 ($\rightarrow$ O2): Transfer learning for cold-start generalization — pretrained embeddings + imbalance-aware training + rigorous split protocols
- Pillar 3 ($\rightarrow$ O3): Intrinsic explainability — attention attribution + Grad-CAM-style visualization + case-study validation
- Pipeline: Data Collection $\rightarrow$ Preprocessing $\rightarrow$ Encoders $\rightarrow$ Fusion $\rightarrow$ Prediction Head $\rightarrow$ Training $\rightarrow$ Evaluation $\rightarrow$ Interpretability

Data Collection — Benchmark Datasets

METHODOLOGY · DATA

Dataset	Scale	Task	Key Feature
BindingDB	Large	DTI / DTA	Extensive Ki, Kd, IC50 affinity data across diverse targets
DAVIS	Medium	DTA	Kinase-focused affinity regression benchmark
KIBA	Medium	DTA	Composite KIBA-score affinity regression benchmark
DrugBank	Supplementary	DTI / pretraining	Generalizability testing and self-supervised pretraining source
ChEMBL	Large	DTI & DTA	Comprehensive bioactivity database; supports both tasks
Luo_17	Large	DTI (binary)	Heterogeneous network with drug-drug

Preprocessing I — Drug Representation

METHODOLOGY · DATA

- SMILES strings parsed with RDKit; duplicates removed; molecules mapped to dataset identifiers
- Atom (node) features: atomic number, degree, hybridization, aromaticity, formal charge
- Bond (edge) features: bond type, conjugation, ring membership
- Pre-trained ChemBERTa embeddings (seyonec/ChemBERTa-zinc-base-v1) fused in as additional node/molecule features for transfer learning

$$\mathcal{G}_d = (V, E), \qquad V = \text{atoms}, \; E = \text{bonds}$$

- FASTA amino-acid sequences tokenized and embedded via pre-trained ESM-2 (facebook/esm2_t6_8M_UR50D, 320-dim; 650M/1280-dim variant for Luo_17)
- Residue-level contextual embeddings capture local chemistry and long-range evolutionary dependencies from large-scale unsupervised pretraining
- Sequences padded/truncated to a fixed maximum length (e.g., 1200 residues) for batch processing
- Two operating modes: full residue sequence (B,L,320) for the BiGRU path, or mean-pooled (B,320) for a lightweight linear-projection path

Preprocessing III — Labels & Data Splitting

METHODOLOGY · DATA

- Positive labels = experimentally validated interactions; negatives via unknown-negative sampling to minimize false-negative contamination
- Stratified negative sampling ratios (e.g., 1:1, 1:10) + imbalance-aware losses (weighted BCE, focal loss)
- Regression targets standardized via -log10 transform: Kd/Ki/IC50 → pKd/pKi/pIC50
- Warm-start: random 10-fold CV · Cold-drug: drug-disjoint · Cold-target: target-disjoint · Cold-pair: pair-disjoint · Cross-dataset: train on one benchmark, test on another

H-GNN-RNN-DTI — System Overview

METHODOLOGY · ARCHITECTURE

- Symmetric dual-branch network: structural drug branch ‖ sequential protein branch
- Component 1 — Modality-specific encoders: Hierarchical GAT (drug) and BiGRU (protein)
- Component 2 — Gated Cross-Attention Fusion: dynamic, bidirectional drug↔protein interaction
- Component 3 — Task-specific prediction heads: DTI classification (sigmoid/softmax) and DTA regression (linear)
- End-to-end differentiable — trained jointly, not as separately-tuned stages

Drug Encoder (I) — Graph Attention Message Passing

- Node initialization fuses physicochemical descriptors with ChemBERTa semantics
- Hierarchical GAT: 3 stacked layers, 8 attention heads per layer
- At each layer, every atom aggregates a learned, attention-weighted message from its bonded neighbors

$$h_i^{(0)} = W_0 \left[x_i \,\|\, c_i \right] + b_0$$

$$e_{ij}^{(l,k)} = \mathrm{LeakyReLU}\Big(a_k^{\top} \left[W_k h_i^{(l)} \,\|\, W_k h_j^{(l)} \,\|\, W_e b_{ij} \right] \Big)$$

$$\alpha_{ij}^{(l,k)} = \frac{\exp\big(e_{ij}^{(l,k)}\big)}{\sum_{u \in \mathcal{N}(i)} \exp\big(e_{iu}^{(l,k)}\big)}$$

$$h_i^{(l+1)} = \Big\Vert_{k=1}^{K} \sigma\left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \alpha_{ij}^{(l,k)} W_k^{(l)} h_j^{(l)} \right)$$

Drug Encoder (II) — Hierarchical Readout

- After L=3 message-passing layers, atom-level states are aggregated into one molecule-level vector
- Mean pooling captures overall composition; max pooling captures the single most salient substructure
- Both are concatenated with a molecule-level ChemBERTa embedding before a final MLP projection

$$z_{\text{mean}} = \frac{1}{|V|} \sum_{i \in V} h_i^{(L)}$$

$$z_{\max} = \max_{i \in V} h_i^{(L)}$$

$$z_d = \text{MLP}\left([z_{\text{mean}} \,\|\, z_{\max} \,\|\, c_{\text{mol}}] \right)$$

Protein Encoder (I) — Bidirectional GRU

- Input: ESM-2 contextual embeddings per residue, x_t $\in \mathbb{R}$^320 (or $\mathbb{R}$^1280 $\rightarrow$ projected for Luo_17)
- 2 stacked layers, forward and backward GRU passes combined at every position
- Captures both N-to-C and C-to-N contextual dependencies — binding-relevant residues can depend on either direction

$$r_t = \sigma\left(W_r x_t + U_r \overrightarrow{h}_{t-1} + b_r\right)$$

$$u_t = \sigma\left(W_u x_t + U_u \overrightarrow{h}_{t-1} + b_u\right)$$

$$\hat{h}_t = \tanh\left(W_h x_t + U_h(r_t \odot \overrightarrow{h}_{t-1}) + b_h\right)$$

$$\overrightarrow{h}_t = (1 - u_t) \odot \overrightarrow{h}_{t-1} + u_t \odot \hat{h}_t \qquad \text{(backward } \overleftarrow{h}_t \text{ symmetric)}$$

$$h_t = \left[\overrightarrow{h}_t \,\|\, \overleftarrow{h}_t\right]$$

Protein Encoder (II) — Attention Pooling

- All T residue-level hidden states are combined into one protein-level vector
- A learned attention vector scores each position's relevance before weighted summation
- Down-weights padding/low-information residues; up-weights residues near probable binding regions

$$u_t = \tanh(W_p h_t + b_p)$$

$$\beta_t = \frac{\exp(u_t^\top v_p)}{\sum_{\tau=1}^{T} \exp(u_\tau^\top v_p)}$$

$$z_p = \sum_{t=1}^{T} \beta_t \, h_t$$

Gated Cross-Attention (I) — Bidirectional Cross-Modal Attention

METHODOLOGY · FUSION

- Core architectural contribution — replaces static concatenation with dynamic, learned interaction
- Atom-level states H_d $\in \mathbb{R}^{\wedge}(n \times d)$ and residue-level states H_p $\in \mathbb{R}^{\wedge}(m \times d)$ attend to each other directly
- Scaled dot-product attention, computed in both directions simultaneously

$$Q_d = H_d W_Q^d, \quad K_p = H_p W_K^p, \quad V_p = H_p W_V^p$$

$$A_{d \to p} = \mathrm{softmax}\left(\frac{Q_d K_p^\top}{\sqrt{d_k}} \right) V_p$$

$$Q_p = H_p W_Q^p, \quad K_d = H_d W_K^d, \quad V_d = H_d W_V^d$$

$$A_{p \to d} = \mathrm{softmax}\left(\frac{Q_p K_d^\top}{\sqrt{d_k}} \right) V_d$$

Gated Cross-Attention (II) — Adaptive Gating

METHODOLOGY · FUSION

- A sigmoid gate learns, per feature dimension, how much cross-modal signal to trust vs. preserve
- Prevents the cross-attention output from overwriting informative but modality-specific features
- Applied symmetrically to both drug and protein branches

$$
\begin{aligned}
g_d &= \sigma\left(W_g^d \left[H_d \,\|\, A_{d\to p}\right] + b_g^d\right) \\
H'_d &= g_d \odot H_d + (1 - g_d) \odot A_{d\to p} \\
g_p &= \sigma\left(W_g^p \left[H_p \,\|\, A_{p\to d}\right] + b_g^p\right) \\
H'_p &= g_p \odot H_p + (1 - g_p) \odot A_{p\to d}
\end{aligned}
$$

Global Fusion & Joint Representation

- Gated atom/residue states are pooled back to single drug and protein vectors
- A second, global-level gate fuses the two into one joint drug-target representation

$$z'_d = \mathrm{READOUT}(H'_d), \qquad z'_p = \mathrm{READOUT}(H'_p)$$
$$g = \sigma\big(W_f\,[z'_d \,\|\, z'_p] + b_f\big)$$
$$z = g \odot z'_d + (1 - g) \odot z'_p$$

Prediction Head — Multi-Task Output

- Shared MLP trunk with dropout and residual connections for stability
- Task-specific output layer: classification head (DTI) or regression head (DTA)
- Same underlying architecture serves both prediction tasks in the study scope

$$h_1 = \mathrm{Dropout}\big(\mathrm{ReLU}(W_1 z + b_1)\big), \qquad h_2 = \mathrm{ReLU}(W_2 h_1 + b_2)$$

$$\text{DTI:} \quad \hat{y} = \mathrm{softmax}(W_o h_2 + b_o) \in \mathbb{R}^2$$

$$\text{DTA:} \quad \hat{y} = W_o h_2 + b_o \in \mathbb{R}$$

Training Strategy (I) — Self-Supervised Pretraining

METHODOLOGY · TRAINING

- Drug and protein encoders optionally pretrained on large unlabeled corpora (e.g., DrugBank) before DTI fine-tuning
- Contrastive learning and masked-feature prediction objectives learn transferable structure without requiring interaction labels
- Directly targets Objective 2 — cold-start generalization to drugs/targets with few or no labeled interactions

$$\mathcal{L}_{\text{contrastive}} = -\log \frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\sum_j \exp(\text{sim}(z_i, z_j)/\tau)}$$

Training Strategy (II) — Imbalance-Aware Losses

METHODOLOGY · TRAINING

- Focal loss down-weights easy negatives, focusing gradient on hard/ambiguous pairs
- Weighted BCE offered as an alternative/ablation baseline for classification
- DTA regression trained with standard MSE on the standardized affinity targets

$$\mathrm{FL}(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t)$$

$$\mathcal{L}_{\mathrm{wbce}} = -\left[w_1 \, y \log(\hat{y}) + w_0 (1 - y) \log(1 - \hat{y}) \right]$$

$$\mathcal{L}_{\mathrm{mse}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

Training Strategy (III) — Optimization & Regularization

METHODOLOGY · TRAINING

- AdamW optimizer with decoupled weight decay, trained end-to-end
- Cosine learning-rate schedule with warmup; early stopping on validation loss
- Dropout regularization throughout; 10-fold cross-validation + held-out test set
- Ablation studies isolate each component: GNN-only, RNN-only, concatenation-fusion, no-pretraining variants

$$\theta^* = \arg\min_{\theta} \; \mathbb{E}_{(x,y)\sim\mathcal{D}}\big[\mathcal{L}(f_\theta(x), y)\big] + \lambda\|\theta\|^2$$

$$\eta_t = \eta_{\min} + \tfrac{1}{2}(\eta_{\max} - \eta_{\min})\big(1 + \cos(t\pi/T)\big)$$

- Classification (DTI): threshold-independent metrics preferred given class imbalance
- Regression (DTA): standard error and correlation metrics on the continuous affinity scale

$$\text{AUC-ROC} = \int_0^1 \text{TPR}\, d(\text{FPR}), \qquad \text{AUPR} = \int_0^1 \text{Precision}\, d(\text{Recall})$$

$$\text{RMSE} = \sqrt{\tfrac{1}{N}\sum (y_i - \hat{y}_i)^2}, \qquad \text{MAE} = \tfrac{1}{N}\sum |y_i - \hat{y}_i|$$

$$\text{PCC} = \frac{\sum (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum (y_i - \bar{y})^2 \sum (\hat{y}_i - \bar{\hat{y}})^2}}$$

Evaluation Protocol (II) — Cold-Start Splits

METHODOLOGY · EVALUATION

Setting	Definition	What it Tests
S1 — Warm-start	Random split, 10-fold CV; drugs & targets may repeat across train/test	Baseline performance (upper bound)
S2 — Cold-drug	Drug-disjoint split; test drugs never seen in training	Generalization to novel compounds
S3 — Cold-target	Target-disjoint split; test proteins never seen in training	Generalization to novel targets
S4 — Cold-pair	Pair-disjoint split; individual drugs/targets may repeat, but not the pair	Generalization to novel interaction patterns

Evaluation Protocol (III) — Ablation & Statistical Rigor

METHODOLOGY · EVALUATION

- Ablation variants: GNN-only, RNN-only, concatenation-fusion, no-pretraining, alternative pooling strategies
- Cross-dataset transfer: train on one benchmark (e.g. BindingDB), evaluate on another (e.g. DAVIS)
- Every configuration run across multiple random seeds; report mean $\pm$ standard deviation
- Paired significance testing (e.g. paired t-test) against baselines — not point-estimate comparison alone

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \qquad d_i = \text{metric}_{\text{hgnn},i} - \text{metric}_{\text{baseline},i}$$

Interpretability & Explainability Framework

- Attention-based attribution: cross-attention weights double as an atom/residue importance signal, at no extra training cost
- Visualization: heatmaps + graph-based Grad-CAM overlays on molecular graphs and protein sequences
- Validation: predicted binding hotspots compared against known crystal structures (Protein Data Bank) in case studies

$$I(v) = \sum_{r} A_{d\to p}[v, r], \qquad I(r) = \sum_{v} A_{d\to p}[v, r]$$

$$L_c^{\text{Grad-CAM}} = \mathrm{ReLU}\Big(\sum_{k} \alpha_k^c A^k\Big), \qquad \alpha_k^c = \frac{1}{Z}\sum_{i,j} \frac{\partial y^c}{\partial A_{ij}^k}$$

Expected Outcomes & Contributions

OUTCOMES

- A robust hybrid GNN-RNN model with a dynamic, gated cross-modal fusion mechanism (methodological contribution)
- Statistically significant predictive improvement over GSRF-DTI, TTModel, DTKGIN and single-modality baselines, including in cold-start settings (empirical contribution)
- An intrinsically interpretable DTI framework producing biologically meaningful, case-study-validated explanations (translational contribution)
- A fully reproducible pipeline — public source code, trained models, and experimental configurations (open-science contribution)

Research Timeline (Illustrative)

OUTCOMES

- Phase 1 — Literature synthesis, data acquisition & preprocessing pipeline, baseline reproduction
- Phase 2 — H-GAT + BiGRU encoders implemented and unit-validated independently ($\rightarrow$ Objective 1 groundwork)
- Phase 3 — Gated Cross-Attention fusion integrated; full-model training, ablations, cold-start evaluation ($\rightarrow$ Objectives 1 & 2)
- Phase 4 — Interpretability layer, case-study validation, cross-dataset transfer experiments ($\rightarrow$ Objective 3)
- Phase 5 — Manuscript preparation, thesis writing, and defense

Summary — Key Takeaways

SUMMARY

- Gap: cross-modal fusion is shallow, cold-start generalization is weak, interpretability is absent
- Response: H-GNN-RNN-DTI — H-GAT drug encoder + BiGRU protein encoder + novel Gated Cross-Attention fusion
- Rigor: cold-start/cross-dataset evaluation, multi-seed statistical testing, ablation-isolated claims
- Payoff: measurable accuracy gains, principled generalization, and biologically grounded interpretability

Thank You

Questions & Discussion

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